Stability of LLM-Based Essay Scoring Under Input Reordering and Few-Shot Prompting

Arnav Palkhiwala*, Maxx Ferrian*, Mohit Mohanraj*

*University of Washington

Keywords: Large Language Model (LLM), Zero-Shot Prompting, Few-Shot Prompting, Input Ordering

Abstract

The reliability of large language models (LLMs) as automated graders depends on whether their evaluations remain stable under changes in prompt structure and input ordering. While previous work has demonstrated that LLM predictions can be sensitive to input order in structured, unambiguous tasks, it is unclear whether similar effects occur in more complex evaluation settings such as essay grading.

In this study, we investigate how prompting strategy and document order influence LLM-based essay scoring. Our results show that prompting strategy and document order produce small but statistically significant shifts in assigned scores. Few-shot prompting generally results in slightly lower scores, while reversed document order leads to modest score increases on the same essays. Despite these effects, most score variability is explained by differences in essay quality, indicating that LLM grading remains largely stable across prompt configurations.

1 Introduction

With the growing capabilities of large language models (LLMs) to process and evaluate large volumes of data, educators and reviewers are increasingly turning to these tools to reduce grading workload and generate scores efficiently. In academic settings, including classroom assessments, college admissions, and research conference reviews, LLMs offer substantial gains in speed and scalability. While automated grading significantly improves efficiency, important questions arise regarding the quality and reliability of these evaluations, particularly whether models produce consistent outputs across runs. The stakes of these determinations are high, as they directly influence academic outcomes, admissions decisions, and conference acceptances, which are contexts where fairness and consistency are critical. Ensuring that LLM-based grading systems are stable and robust is therefore essential before broader deployment in high-impact decision-making environments.

Previous literature demonstrates that LLMs are sensitive to input ordering in structured tasks. In multiple-choice settings, simply reordering answer options has been shown to produce substantial performance gaps, ranging from **13–75%** [1]. This finding highlights the extent to which LLMs exhibit positional bias when selecting among a list of options. Related work further shows that reshuffling inputs across tasks such as paraphrasing and relevance judgment can also degrade performance, and while few-shot prompting may reduce this sensitivity, it does not eliminate the effect entirely [2]. Studies comparing zero-shot and few-shot prompting in specialized domains, such as in biotech for medical clinics, indicate that model architecture and prompting strategy both play a significant role in performance quality [3]. However, most prior work focuses on classification, reasoning, or option-selection tasks. Compared to these settings, essay grading represents a more holistic and complex evaluation problem. Essays require longer contextual understanding, more nuanced reasoning, and the assignment of a continuous score rather than selection from fixed alternatives. As a result of the entire evaluative and scoring responsibility resting with the LLM, determining whether order-related biases persist in such complex judgment tasks is crucial.

In this paper, we extend prior work by investigating whether the order in which essays are presented within a grading batch affects the score assigned to each essay. Specifically, we examine scenarios in which multiple essays are evaluated simultaneously and test whether reordering the same set of essays leads to changes in assigned grades. In addition to measuring order effects, we analyze how prompting strategy influences model stability by comparing zero-shot prompting with few-shot prompting. Our examples for this include essays representing exceptional, average, and poor quality writing, and we will also identify scoring variance based on quality level. We define consistency as the stability of assigned scores when identical essays are presented in different input sequences, with or without supporting demonstrations. While previous research has documented order sensitivity in classification and reasoning tasks, its implications for holistic evaluation tasks such as automated essay grading remain underexplored. We provide a systematic analysis of input-order sensitivity in LLM-based automated grading under both zero-shot and few-shot prompting conditions.

2 Experimental Design

To evaluate the consistency and potential biases of LLM-based grading under different prompting configurations, we designed a controlled experimental framework that systematically varies both document order and prompting strategy. The primary objective of this study is to assess whether grading outcomes differ when documents are presented in ascending versus descending order and when the model is prompted using zero-shot versus few-shot techniques. All experiments were conducted using the **Gemini-3-Flash-Preview** model (using temperature 1.0), accessed via the Gemini API under a fixed inference configuration [4].

Documents were randomly sampled from our entire database (not based on quality type) without replacement in batches of five at a time. Sampling without replacement ensured that each document was evaluated only once within a given experimental cycle, thereby preventing duplication and preserving independence across observations. This batch-based sampling procedure was repeated iteratively until all documents in the dataset had been evaluated.

The dataset was constructed to span a broad range of writing quality while maintaining diversity in content and complexity. Poor quality essays consisted of K–6 narrative writing samples [5], [6]. Average quality essays included 10th–12th grade and undergraduate-level essays drawn from English courses and college application writing [6], [7], [8]. Exceptional quality essays were drawn from introductions and conclusions of research papers in science and technology domains sourced from arXiv [9]. Our dataset consists of **100** examples for each essay quality type, giving us **300** essays that are being scored in total. This stratified sampling approach ensured clear variation in writing proficiency across the evaluation set.

Each sampled document was graded by the LLM under four distinct experimental conditions. Specifically, every document received a score under ascending order with zero-shot prompting, descending order with zero-shot prompting, ascending order with few-shot prompting, and descending order with few-shot prompting. In all conditions, the model assigned a numerical grade on a scale from **1 to 10**, where 1 represented the lowest performance and 10 represented the highest performance according to the evaluation criteria defined in the prompt. The grading criteria focuses on writing quality over content as we did not want specific topics to be confounded with higher scores. Instead, the primary goal is simply looking at writing flow scores and if they remain the same under different conditions.

The ascending and descending order conditions refer to the sequence in which documents were presented to the model within a batch, allowing us to test for potential order effects in grading behavior. Ascending is simply the randomized order, while descending represents the reverse of this same randomized order. The zero-shot condition provided only task instructions, while the few-shot condition included example graded documents intended to guide the model's evaluation. The few-shot prompt included one exemplar essay from each quality category (poor, average, and exceptional) to provide representative calibration across the writing spectrum.

LLM Grading Prompt (Abbreviated/Summarized — Not Verbatim from the Real Prompt)

```
You are an instructor grading different
essays.

Grade holistically using these dimensions:
flow, transitions, content quality and
focus, spelling and grammar, knowledge and
depth, and structure.

Provide one overall integer score from 1 to
10, where 1 indicates fundamentally flawed
writing, 5 indicates mixed or adequate
writing with clear weaknesses, and 10
indicates exemplary writing with no
meaningful weaknesses.  Avoid score
inflation.

In the few-shot condition, the prompt
additionally includes one poor, one
average, and one exceptional essay as
calibration anchors.

The model is instructed to return a JSON
array with one score and justification per
essay, preserving the input order exactly.
```

See Appendix 6 for the full prompt text.

For each document, we recorded its unique identifier along with the four grades assigned under the different experimental conditions. The LLM outputs for each batch is stored in JSON format file containing **document ID, score, and a short justification**. All results were then stored in a structured dataset indexed by document ID and condition, enabling direct within-document comparisons across prompting strategies and ordering configurations. Because each document was evaluated under all four conditions, the design supports paired statistical analyses that control for document-level variability.

The final dataset consists of four numerical scores per document, corresponding to the four prompting conditions. These data serve as the basis for subsequent statistical analysis, including paired comparisons and variance analyses, to determine whether prompting strategy, document order, or their interaction significantly influences grading outcomes.

The full implementation of the grading pipeline, dataset construction scripts, and analysis code are publicly available at: `https://github.com/ArnavPalkhiwala/STAT496Capstone`.

3 Results and Analysis

This section examines whether LLM grading is influenced by (1) prompting strategy (few-shot vs. zero-shot), (2) presentation order (ascending/original vs. descending/reversed), and (3) the objective serial position of an essay within a five-essay grading batch. Because each essay was graded multiple times and within structured five-essay groups, linear mixed-effects models were used to account for repeated measurements at both the essay level and the batch level.

3.1 Descriptive Statistics

Across all 1,200 observations, mean scores varied modestly across conditions. In the earlier order-based analysis, zero-shot prompting generally produced slightly higher scores than few-shot prompting, and reversed (descending) presentation produced slightly higher scores than original (ascending) presentation. These same general patterns were reflected in the batch-level means, indicating that the effects were not limited to individual essay-level noise.

3.2 Mixed-Effects Model: Observation Level (Order Analysis)

To account for repeated grading of essays and shared context within five-essay groups, we fit a linear mixed-effects model with random intercepts for essay ID and batch group. Fixed effects included prompting strategy, order, and their interaction. *The full results shown in the following can be found in the appendix 6.1.*

Results indicated a significant ($\alpha \leq 0.05$) main effect of prompting strategy,

$$F(1, 944.59) = 41.91, \quad p < .001,$$

with few-shot prompting producing lower scores than zero-shot prompting (estimated difference ≈ -0.35 points).

There was also a significant main effect of presentation order,

$$F(1, 944.59) = 13.49, \quad p < .001,$$

with reversed presentation producing higher scores than original order (estimated difference $\approx +0.20$ points).

The interaction between prompting strategy and order was not significant,

$$F(1, 944.59) = 0.39, \quad p = .534,$$

indicating that the prompting effect does not depend on presentation order.

Variance components showed substantial between-essay variability and relatively little batch-level variance, suggesting that most score variability is attributable to stable differences in essay quality rather than experimental context.

3.3 Mixed-Effects Model: Batch Level (Order Analysis)

To evaluate whether order and prompting effects remain when collapsing each grading run into a single outcome, we fit a second mixed-effects model using the mean score of each five-essay batch as the dependent variable, with a random intercept for batch group.

The results replicated the observation-level findings. There was a significant main effect of prompting strategy,

$$F(1, 177) = 22.45, \quad p < .001,$$

and a significant main effect of order,

$$F(1, 177) = 7.22, \quad p = .008.$$

The interaction was not significant,

$$F(1, 177) = 0.21, \quad p = .649.$$

Thus, both prompting strategy and presentation order produce modest but reliable shifts in average scores at the grading-session level.

3.4 Objective Serial Position Analysis

While ascending versus descending order captures directional framing, it does not directly test whether essays appearing earlier or later within a grading sequence receive systematically different scores. To evaluate potential primacy, recency, or fatigue effects, we modeled objective position (0–4 within each five-essay batch). These models included random intercepts for essay ID and batch group, although the batch-group variance was estimated to be effectively zero in the updated models.

3.4.1 Linear Position Model

Objective position was first modeled as a continuous predictor:

$$\text{Score} \sim \text{mode} \times \text{position} + (1|essay) + (1|group).$$

Under multiple testing, the model showed a small but statistically significant main effect of prompting strategy,

$$F(1, 897) = 3.96, \quad p = .047,$$

with few-shot prompting associated with slightly higher scores in this specification (estimate $= 0.213$).

However, there was no significant linear effect of position,

$$F(1, 897) = 2.56, \quad p = .110,$$

and no significant interaction between prompting strategy and position,

$$F(1, 897) = 0.04, \quad p = .849.$$

Thus, there is no evidence of a consistent linear increase or decrease in scores as essays appear later within a five-essay grading run.

3.4.2 Nonlinear Position Model

To test for potential nonlinear patterns, such as first-position or last-position advantages, position was also treated as a categorical factor:

$$\text{Score} \sim \text{mode} \times \text{position}_{factor} + (1|essay) + (1|group).$$

In this model, prompting strategy again showed a significant main effect,

$$F(1, 893) = 10.06, \quad p = .002.$$

The overall main effect of position approached, but did not reach, conventional statistical significance. Thus, overall, we can say it is not statistically significant,

$$F(4, 458.73) = 2.37, \quad p = .052,$$

and the interaction between prompting strategy and position was not significant,

$$F(4, 893) = 0.22, \quad p = .926.$$

At the coefficient level, relative to position 0 in the zero-shot condition, essays in position 1 were scored significantly lower ($\beta = -0.842$, $p = .013$), and essays in position 3 were also scored significantly lower ($\beta = -0.900$, $p = .008$). However, because the omnibus factor test for position was only marginal and the Tukey-adjusted pairwise comparisons within each prompting condition were not statistically significant, these position-specific differences should be interpreted cautiously.

More broadly, the post-hoc comparisons showed no reliable pairwise differences among positions within either the zero-shot or few-shot conditions after multiple-comparison adjustment. This indicates that any apparent position-specific deviations are not robust enough to support a clear primacy, recency, or fatigue interpretation.

For complete statistical details of the experimental results, refer to the appendix (Table 1 and Table 2).

4 Interpretation and Discussion

4.1 Influence of Prompting Strategy

Across the analyses, the prompting strategy remained one of the most consistent factors influencing the scores assigned by the model, with different strategies leading to different scores on the same essays. In the primary order-based models, few-shot prompting produced lower scores than zero-shot prompting.

$$F(1, 944.59) = 41.91, \quad p < .001$$

The estimated difference between conditions was approximately 0.35 points. The batch-level analysis replicated this pattern:

$$F(1, 177) = 22.45, \quad p < .001$$

These results suggest that including calibration examples in the prompt changes how the model interprets the scoring scale.

One plausible explanation is that the few-shot examples act as anchor points that limit the model's interpretation of the rubric. When the model is provided with explicit examples representing poor, average, and strong writing, it may evaluate new essays relative to these anchors and therefore apply the scoring rubric more conservatively. In contrast, the zero-shot configuration relies entirely on textual instructions describing the rubric, which may allow the model more flexibility in interpreting what constitutes a higher score.

However, the serial position models provide a more nuanced perspective. In the linear position model, the prompting strategy showed a smaller effect and in the opposite direction:

$$F(1, 897) = 3.96, \quad p = .047$$

In this specification, few-shot prompting was associated with slightly higher scores. This suggests that the influence of the prompting strategy may depend on how the surrounding context is modeled. While the main order-based analysis indicates a downward shift under few-shot prompting, the position-based models indicate that this effect is relatively small and sensitive to modeling assumptions. Taken together, the results suggest that the prompting strategy influences score calibration but does not dramatically alter the relative ranking of essays.

4.2 Influence of Presentation Order

Presentation order also produced a statistically significant effect on the scores assigned by the model.

$$F(1, 944.59) = 13.49, \quad p < .001$$

Essays evaluated in reversed order received slightly higher scores on average than essays evaluated in the original order, with the mixed-effects model estimating an increase of approximately 0.20 points. The batch-level model confirmed that this effect persists when analyzing grading sessions as aggregated outcomes:

$$F(1, 177) = 7.22, \quad p = .008$$

Although the magnitude of this shift is small relative to the full ten-point scoring scale, the consistency of the effect suggests that the contextual structure of the prompt can influence evaluation behavior. When multiple essays are presented within the same prompt, the model processes them within a shared context window. As a result, the interpretation of one essay may be influenced by the surrounding essays in the sequence. Reversing the order of essays changes the contextual comparisons that the model implicitly performs during evaluation.

Importantly, there was no significant interaction between prompting strategy and presentation order, indicating that the order effect occurs independently of whether calibration examples are included in the prompt.

4.3 Serial Position Within Grading Batches

While the ascending versus descending manipulation captures a general ordering effect, it does not directly test whether essays appearing earlier or later in a grading batch receive systematically different scores. To address this question, we modeled the objective position of essays within the five-essay grading sequence.

The linear position model found no significant relationship between position and score:

$$F(1, 897) = 2.56, \quad p = .110$$

This suggests that scores do not consistently increase or decrease as essays appear later in a grading sequence, providing no evidence for fatigue effects, primacy advantages, or recency advantages across the five positions.

The categorical position model produced a slightly more complex pattern. The overall effect of position approached statistical significance:

$$F(4, 458.73) = 2.37, \quad p = .052$$

However, these differences were not supported by the omnibus factor test and did not remain significant after Tukey-adjusted multiple comparisons. As a result, these position-specific deviations should be interpreted cautiously.

Overall, the position analyses suggest that while small fluctuations in scores may occur at specific positions, there is no consistent evidence of a systematic serial position effect.

4.4 Relative Importance of Essay Quality

The variance structure of the mixed-effects models provides additional insight into the sources of score variation. The models show that most variability occurs at the essay level rather than at the batch or prompt configuration level. This indicates that intrinsic differences in essay quality explain the majority of score variation, while experimental factors such as prompting strategy and ordering contribute relatively small adjustments.

This finding is encouraging from the perspective of automated grading systems. Even though prompt configuration can produce measurable shifts in average scores, the magnitude of these effects remains modest compared to the variability introduced by differences in writing quality.

4.5 Implications for LLM-Based Evaluation

Taken together, the results suggest that LLM-based essay grading systems are generally stable but not completely insensitive to prompt structure. Prompting strategy can influence how the scoring rubric is interpreted, while presentation order introduces small contextual framing effects when grading multiple documents simultaneously. However, the dominant factor determining scores remains the intrinsic quality of the essay.

From a practical perspective, these findings highlight several considerations for deploying LLM-based evaluation systems. Calibration examples in few-shot prompts may influence the overall scoring scale, potentially helping to standardize rubric interpretation. Additionally, because presentation order introduces small but measurable shifts in scores, grading pipelines may benefit from randomizing or standardizing document order to reduce contextual bias.

Overall, the results indicate that LLM-based grading systems exhibit encouraging stability across different prompting configurations, but prompt design choices still play a role in shaping model behavior. Understanding these contextual influences will be increasingly important as LLMs are adopted for automated evaluation tasks in educational and professional settings.

5 Limitations

While our results and process are thorough, there are a few areas in which our analysis could be enriched.

Type of data used. We have essays that come from a variety of different sources: elementary school, middle/high school, college undergraduate, and research papers. While this diversity provides a strong testing ground to evaluate LLM grading performance across a wide range of writing styles and topics, the essays do not focus on a single subject area. This could increase the variability of the LLM's evaluations because the judging criteria may slightly differ from essay to essay. Our intention was to ensure that grading was based primarily on writing quality and flow rather than genre or writer age. Additionally, the examples used in few-shot prompting could bias the model toward essays that are semantically similar to those anchor essays. However, the main metric we measure is how the grading of an individual essay changes under different experimental conditions, meaning each essay is evaluated relative to itself across runs. Overall, this factor did not have a major negative impact on the experiment.

Quantity of data and batch size. Our experiment consists of 300 essays due to the difficulty in sourcing high-quality examples without generating synthetic data, as well as the compute costs associated with running our pipeline at a larger scale. We also used a batch size of five essays at a time. In real-world scenarios, however, LLM grading systems may need to process significantly larger batches of essays in each iteration, which could potentially alter the results.

LLM choice. The LLM we used for evaluation is Gemini-3-Flash-Preview. However, as is widely recognized in the LLM research community, different models can behave and perform differently. Other models may exhibit less sensitivity to prompt structure or input reordering. Due to compute constraints, we were only able to evaluate a single model in this study.

Lack of ground truth grading. Our experiment evaluates the consistency of LLM-generated scores rather than their absolute accuracy. Because the dataset does not include human-annotated ground truth grades for each essay, we cannot determine whether the scores assigned by the model are correct. Instead, our analysis focuses on whether the same essay receives different scores when input order or prompting conditions change.

6 Conclusion

Our findings suggest that in the current LLM ecosystem, models are capable of acting as effective graders when provided with a thorough grading rubric. In our work, we use 300 essays ranging from elementary school student essays to college essays and published science and technology research papers. We split the essays into batches of size 5 using a random sample from the 300 essays (without replacement) and evaluate the grading for each individual paper under four conditions: zero-shot prompting with regular essay order, zero-shot prompting with reversed essay order, few-shot prompting with regular essay order, and few-shot prompting with reversed essay order.

The results show that prompting strategy impacts the model's interpretation of the scoring rubric, especially when examples are provided, which can lead to a potentially more standardized grading procedure. Essays are typically scored slightly lower when provided with few-shot examples. Randomizing input order also appears to be the most optimal approach to ensure greater objectivity in grading, as reversed order led to slightly higher scores on the same essays.

At a larger scale, these findings suggest that LLMs can be feasible graders. They provide some level of transparency while offering significant improvements in efficiency and reductions in grading time. However, human oversight is still recommended because the stakes, whether a grade for a class or admission into a university or research conference, are high. Even a small inaccuracy could have a meaningful impact on individuals or institutions. Overall, this experiment provides evidence that LLMs have the potential to be used as fair and effective grading tools in the future when deployed with careful prompt design and oversight. Future work could expand this study by testing additional models, larger datasets, and different batch sizes to better understand how prompt structure influences automated grading systems.

Acknowledgments

We would like to thank Anne Wagner for her guidance and comments during this research process.

References

[1] Pouya Pezeshkpour and Estevam Hruschka. Large Language Models Sensitivity to the Order of Options in Multiple-Choice Questions. 2023. https://arxiv.org/abs/2308.11483

[2] Bryan Guan, Tanya Roosta, Peyman Passban, and Mehdi Rezagholizadeh. The Order Effect: Investigating Prompt Sensitivity to Input Order in LLMs. 2025. https://arxiv.org/abs/2502.04134

[3] Yanis Labrak, Mickael Rouvier, and Richard Dufour. A Zero-shot and Few-shot Study of Instruction-Finetuned Large Language Models Applied to Clinical and Biomedical Tasks. 2023. https://arxiv.org/abs/2307.12114

[4] Google DeepMind. Gemini 3 Flash Preview Model Documentation. Accessed March 12, 2026. https://ai.google.dev/gemini-api/docs/models

[5] TTMS. Writing Samples Collection. Accessed March 12, 2026. https://www.ttms.org/PDFs/03%20Writing%20Samples%20v001%20(Full).pdf

[6] Thoughtful Learning. Free Grade-Level Student Writing Models. Accessed March 12, 2026. https://k12.thoughtfullearning.com/resources/studentmodels

[7] New York University. Essay Examples. Accessed March 12, 2026. https://www.nyu.edu/content/dam/nyu/univPrgms/documents/summer/ESSAY%20EXAMPLES.pdf

[8] Johns Hopkins University Admissions. Essays That Worked. Accessed March 12, 2026. https://apply.jhu.edu/hopkins-insider/application-section/essays-that-worked/

[9] arXiv. arXiv.org e-Print Archive. Accessed March 12, 2026. https://arxiv.org

Appendix

Full LLM Grading Prompt

```
You are an instructor grading different essays.

Grade holistically using these dimensions:

• 1) Flow (clarity/readability across sentences)
• 2) Transitions (connections between ideas/paragraphs)
• 3) Content Quality & Focus (relevance, specificity, alignment to the week's concepts)
• 4) Spelling & Grammar (correctness, professionalism)
• 5) Knowledge & Depth (understanding + thoughtful engagement)
• 6) Structure (organization, paragraphing, logical progression)

Score rules:

• Provide ONE overall score from 1 to 10.
• They should be integers only.
• 1 = fundamentally flawed; minimal understanding; very poor writing.
• 5 = mixed/adequate; some understanding; clear weaknesses.
• 10 = exemplary; polished, insightful, well-structured; no meaningful weaknesses.
• Avoid score inflation:  9--10 only if nearly flawless.
```

Few-shot only (inserted before essays):

```
Calibration anchors (use these to calibrate your scoring scale):

[ANCHOR A --- 10/10 EXAMPLE]
{anchor_exceptional}

[ANCHOR B --- 5/10 EXAMPLE]
{anchor_average}

[ANCHOR C --- 1/10 EXAMPLE]
{anchor_poor}
```

Rest of Prompt for All Conditions

```
Now grade EACH essay below independently.

IMPORTANT RULES:

• Return a JSON ARRAY only.
• The array must have EXACTLY {len(essays)} items.
• Items must be in the SAME ORDER as the essays appear below.
• Use the essay ID exactly as provided.
• pred_score must be an integer 1--10.

Essays (in order):
```

```
{essays_block}

Output STRICT JSON ONLY (no markdown, no extra text).

Each object must match:

{"id":"<string>","pred_score":<integer 1-10>,"justification":"<2-4 sentences>"}
```

6.1 Full Experimental Results

Factor	Model Level	Statistic	F-value	p-value	Effect
Prompting Strategy	Observation Level	Mixed Effects	$F(1, 944.59)$	$< .001$	≈ -0.35 score
Presentation Order	Observation Level	Mixed Effects	$F(1, 944.59)$	$< .001$	$\approx +0.20$ score
Prompting Strategy	Batch Level	Mixed Effects	$F(1, 177)$	$< .001$	small decrease
Presentation Order	Batch Level	Mixed Effects	$F(1, 177)$	.008	small increase
Prompting Strategy	Linear Position Model	Mixed Effects	$F(1, 897)$	.047	$+0.213$ score
Essay Position	Linear Position Model	Mixed Effects	$F(1, 897)$	.110	not significant
Prompting Strategy	Nonlinear Position Model	Mixed Effects	$F(1, 893)$	.002	significant
Essay Position	Nonlinear Position Model	Mixed Effects	$F(4, 458.73)$	.052	marginal
Interaction (Mode × Position)	Nonlinear Position Model	Mixed Effects	$F(4, 893)$	.926	none

Table 1. Summary of statistical tests evaluating the effects of prompting strategy, presentation order, and essay position on LLM grading scores.

Prompting	Order	Description
Zero-shot	Ascending	No examples, original essay order
Zero-shot	Descending	No examples, reversed essay order
Few-shot	Ascending	Includes calibration examples
Few-shot	Descending	Calibration examples with reversed order

Table 2. Experimental prompting configurations used in the study.

6.2 Linear Mixed-Effects Model for Position

This model tested whether predicted score changed linearly across essay position in the 5-essay batch and whether that trend differed between zero-shot and few-shot prompting.

Model formula:

`pred_score ~ mode * position + (1 | id) + (1 | group_id)`

Number of observations: 1200

Random-effects groups: id = 300, group_id = 60

Random Effects

Group	Variance	Std. Dev.
id (Intercept)	5.794	2.407
group_id (Intercept)	≈ 0	≈ 0
Residual	1.150	1.072

Fixed Effects

Term	Estimate	Std. Error	df	t value	p value
Intercept	6.9850	0.1583	448.68	44.122	$< .001$
modefewshot	0.2133	0.1072	897.00	1.989	0.047
position	-0.0308	0.0310	897.00	-0.996	0.319
modefewshot:position	-0.0083	0.0438	897.00	-0.190	0.849

Type III ANOVA

Effect	NumDF	DenDF	F value	p value
mode	1	897	3.9580	0.04695
position	1	897	2.5569	0.11017
mode:position	1	897	0.0362	0.84907

95% Confidence Intervals

Term	2.5%	97.5%
Intercept	6.6747	7.2953
modefewshot	0.0032	0.4235
position	-0.0915	0.0298
modefewshot:position	-0.0941	0.0775

Note. The random intercept variance for group_id was approximately zero (singular fit).

6.3 Nonlinear Mixed-Effects Model for Position

This model tested whether the effect of essay position was nonlinear by treating position as categorical.

Model formula:

`pred_score ~ mode * position_f + (1 | id) + (1 | group_id)`

Random Effects

Group	Variance	Std. Dev.
id (Intercept)	5.694	2.386
group_id (Intercept)	≈ 0	≈ 0
Residual	1.154	1.074

Fixed Effects

Term	Estimate	Std. Error	df	t value	p value
Intercept	7.3333	0.2389	386.45	30.698	$< .001$
modefewshot	0.1500	0.1387	893.00	1.082	0.280
position_f1	-0.8417	0.3378	386.45	-2.491	0.013
position_f2	-0.1833	0.4020	345.91	-0.456	0.649
position_f3	-0.9000	0.3378	386.45	-2.664	0.008
position_f4	-0.1250	0.1387	893.00	-0.901	0.368
modefewshot:position_f1	0.0750	0.1961	893.00	0.382	0.702
modefewshot:position_f2	0.1417	0.1961	893.00	0.722	0.470
modefewshot:position_f3	0.0417	0.1961	893.00	0.212	0.832
modefewshot:position_f4	-0.0250	0.1961	893.00	-0.127	0.899

Type III ANOVA

Effect	NumDF	DenDF	F value	p value
mode	1	893.00	10.0558	0.00157
position_f	4	458.73	2.3712	0.05166
mode:position_f	4	893.00	0.2232	0.92554

6.4 Estimated Marginal Means and Pairwise Comparisons

Estimated Marginal Means

Zero-shot:

- Position 0: 7.33
- Position 1: 6.49
- Position 2: 7.15
- Position 3: 6.43
- Position 4: 7.21

Few-shot:

- Position 0: 7.48
- Position 1: 6.72
- Position 2: 7.44
- Position 3: 6.62
- Position 4: 7.33

Pairwise Comparisons

All pairwise comparisons were Tukey-adjusted. No comparisons reached conventional significance thresholds.

Code and Reproducibility

All code used to run the experiments, generate prompts, and perform statistical analysis is available at:
`https://github.com/ArnavPalkhiwala/STAT496Capstone`